

Atmospheric Channel Modelling for Terrestrial SNS-TFQKD under Delhi-like Conditions

Technical Literature Survey and Implementation Briefing

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1 Objective

1.1 Atmospheric Channel Modelling for Terrestrial FSO

Free-space optical (FSO) communication links transmit laser beams through the open atmosphere, making channel quality dependent on local weather and turbulence conditions. The received optical intensity is degraded by three separable physical mechanisms:

1. **Weather-driven attenuation** : fog, haze, and aerosols scatter the beam;
2. **Atmospheric turbulence** : refractive-index fluctuations cause irradiance scintillation;
3. **Pointing errors** : mechanical misalignment and beam wander produce random geometric loss.

These effects can be modelled as multiplicative fading terms on the received irradiance, enabling closed-form or simulation-based performance analysis.

1.2 Relevance to TF-QKD and SNS-TFQKD

Twin-Field QKD protocols, including the Sending-or-Not-Sending (SNS) variant, achieve key-rate distance scaling by interfering single-photon fields at a central relay. For terrestrial free-space deployment, the optical channel between each party and the relay is a turbulent FSO link. Atmospheric transmittance directly determines the photon detection probability (PDTC), which in turn governs:

- The quantum bit error rate (QBER);
- The raw detection gain;
- Ultimately, the secret key rate (SKR).

Accurate atmospheric channel modelling is therefore essential for credible performance prediction. Yu et al. (2022) explicitly modelled SNS-TFQKD through atmosphere-induced PDTC and showed that real-time transmittance selection is necessary to stabilise short-range performance.

1.3 Motivation: Delhi-like Atmospheric Environments

Delhi represents a severe urban FSO scenario due to:

- Dense fog in winter months (November–February);
- Year-round particulate haze driven by vehicular and industrial aerosols;
- Moderate-to-strong optical turbulence over heated urban surfaces.

Studies specific to Delhi (Srivastava et al. 2020; Kesarwani et al. 2018) report visibility falling below 200 m during dense fog events, requiring high link margin designs. This combination of impairments is not adequately captured by models calibrated for European or North American conditions, motivating a locally validated composite channel model.

2 Key Literature Findings

2.1 Atmospheric Turbulence

Rytov variance and turbulence regimes. The Rytov variance (σ_R^2) quantifies turbulence strength and is computed from the refractive-index structure parameter C_n^2 and link length L . It defines the regime boundary:

- $\sigma_R^2 < 0.3$: weak turbulence;
- 0.3–0.5: moderate turbulence;
- $\sigma_R^2 > 5$: strong turbulence.

Raj et al. (2015) reviewed local C_n^2 prediction models, and Pernice et al. (2016) measured σ_R^2 and scintillation index on an indoor weak-turbulence link to validate their Gamma-Gamma model.

Table 1: Turbulence models considered

Model	Validity	Key Feature	Status
Log-Normal	Weak ($\sigma_R^2 < 0.3$)	Single-parameter PDF	Reference only
Gamma-Gamma	Weak–strong	Two-parameter (α, β) PDF	Default model
Nakagami- m	Moderate	Common in MIMO-FSO studies	Not prioritised

Recommendation: Based on the surveyed literature, the Gamma-Gamma distribution is recommended as the default turbulence model, parameterised by α and β derived from σ_R^2 and C_n^2 . The Log-Normal distribution should be retained only as a weak-turbulence sanity check. The 2025 OFC result cited in the synthesis report confirms that log-normal fails to capture irradiance fluctuations even in weak turbulence, strengthening this choice.

2.2 Atmospheric Attenuation

All three major visibility-based fog models follow the Beer-Lambert extinction law with a wavelength-dependent scattering coefficient derived from measured or estimated visibility (V):

Recommendation: The reviewed studies indicate that the Kim model should be implemented first because it is computationally simple and appears directly in both Delhi-specific papers. An upgrade to Al-Naboulsi is advisable if local meteorological data allow fog-type classification (advection vs. radiation). The Kruse model should be used only for benchmarking comparison.

Table 2: Visibility-based attenuation models considered in this study

Model	Fog Type	Key Formula Feature	Recommended Use
Kim	General and dense fog	Simplified visibility exponent $q(V)$; widely validated for terrestrial FSO links	First implementation choice
Kruse	General fog	Classical empirical attenuation model; tends to be less accurate under dense fog conditions	Comparison baseline only
Al-Naboulsi	Advection and radiation fog	Distinguishes fog types; physically motivated attenuation formulation	Use when fog type information is available

2.3 Pointing Errors

Physical origin. Building sway, thermal expansion, and wind loading cause beam wander at the receiver. The resulting misalignment between the beam centroid and detector aperture produces a random intensity fade on top of turbulence fading.

Mathematical modelling. García-Zambrana et al. (2012) established the standard misalignment fading model: the radial pointing displacement is modelled as a Gaussian-distributed jitter with standard deviation σ_s , and the beam footprint at the receiver is approximated by a Gaussian beam of width w_z . The pointing loss is expressed in terms of the error function, yielding a tractable closed-form fading pdf. Bhatnagar and Anees (2015) confirmed this model in a MIMO context and showed significant diversity degradation even under Alamouti coding when pointing errors are present. Borah and Voelz (2009) provided wave-optics-based numerical validation that turbulence can partially mitigate pointing sensitivity by enlarging the effective beam footprint.

Impact on performance. Pointing errors introduce an additional multiplicative loss term that is statistically independent of turbulence scintillation and atmospheric attenuation. They reduce outage capacity and increase BER by narrowing the effective SNR distribution. Multiple papers (Verma 2017; Han 2013; Khodadadi 2017) indicate that omitting pointing errors significantly overestimates FSO link performance.

Recommendation: The Gaussian jitter/misalignment model of García-Zambrana et al. (2012) is recommended as a multiplicative fading block with parameters: beam radius w_z , detector aperture radius a , and jitter standard deviation σ_s .

2.4 FSO-QKD and SNS-TFQKD

Incorporating atmospheric effects. In QKD over FSO, the composite atmospheric loss is collapsed into a time-varying channel transmittance $\eta(t)$, sometimes called the probability distribution of transmittance channel (PDTC). Key references in the attached literature cover this approach:

- Alshaer et al. (2021) combined Gamma-Gamma turbulence and pointing errors in the BB84 protocol and derived closed-form symbol error rate, raw key rate, and secret key rate expressions.
- Yu et al. (2022) modelled free-space SNS-TFQKD with atmosphere-induced PDTC and proposed a prefixed-threshold real-time selection scheme to improve performance.

- Ntanos et al. (2021) demonstrated decoy-state BB84 over a terrestrial FSO link under urban solar radiance conditions.

Common performance metrics across QKD-FSO papers include: QBER, detection gain (Q), raw key rate, and secret key rate (SKR). Background solar photons, detector dark counts, and excess noise are treated as receiver-side noise terms rather than channel model components.

Recommendation: The atmospheric front-end should be modelled as a composite loss (Section 4), then instantaneous transmittance should be mapped to QKD metrics via established expressions for SNS-TFQKD (following Yu et al. 2022). Background noise and detector loss should be implemented as a separate receiver-noise module.

3 Delhi-Specific Atmospheric Findings

The following table consolidates findings from the two primary Delhi-focused sources in the literature set (Srivastava et al. 2020; Kesarwani et al. 2018), supplemented by the general Indian FSO study by Marazi and Raj (2018) and the Bhubaneswar IMD-data study (Padhy et al. 2020) where Delhi-relevant parameters are indicated.

Table 3: Delhi-specific atmospheric parameters used for FSO-QKD modelling

Parameter	Value / Condition	Season / Context	Source
Dense fog visibility	< 200 m (sometimes < 50 m)	Winter (Nov–Feb)	Kesarwani (2018); Srivastava (2020)
Light fog visibility	200–1000 m	Early winter mornings	Srivastava (2020)
Haze visibility	1–4 km	Year-round (pollution driven)	Srivastava (2020)
Clear-sky visibility	> 10 km	Post-monsoon and summer	Srivastava (2020)
Recommended Tx power (fog, 6–8 km)	25–35 dBm for error-free link	Dense fog months	Kesarwani (2018)
Attenuation (dense fog)	Up to 350 dB/km (estimated)	Winter fog events	Derived using Kim model for $V < 200$ m
Attenuation (haze)	~ 10 –30 dB/km	Pollution haze conditions	Srivastava (2020)
Attenuation (clear sky)	< 2 dB/km	Clear conditions	Marazi and Raj (2018)
Aerosol/dust effect	Elevated extinction; PM _{2.5} dominant	Year-round; strongest in winter	Srivastava (2020)
Turbulence strength C_n^2	10^{-13} – 10^{-15} m ^{-2/3}	Urban near-surface atmosphere	Raj et al. (2015)
Turbulence regime	Moderate to strong	Summer afternoons over urban asphalt	Raj et al. (2015)
Dominant availability threat	Dense winter fog (rather than rain)	Nov–Feb	Kesarwani (2018); Srivastava (2020)
Recommended wavelength	1550 nm preferred	All seasons	Padhy (2020); Marazi and Raj (2018)

Key takeaway for Delhi: The surveyed literature suggests that the link budget must be designed for dense winter fog as the worst-case availability scenario. A Kim-model attenuation block parameterised by seasonal visibility distributions ($V < 200$ m for 10–20 days/year in dense fog) is the appropriate starting point. Turbulence is a secondary concern during fog months (low solar heating) but becomes significant in summer. Pointing error is a year-round concern for building-mounted FSO.

4 Recommended Atmospheric Channel Architecture

The recommended composite model is:

$$h_{\text{total}} = h_{\text{atm}} \times h_{\text{GG}} \times h_{\text{point}}$$

Table 4: Components of the composite atmospheric channel

Component	Physical Meaning	Literature Support	Advantages	Limitations
h_{atm} (Atmospheric attenuation)	Deterministic mean path loss from weather (fog, haze, rain). Modelled as Beer-Lambert exponential decay with Kim/Al-Naboulsi visibility-based extinction coefficient.	Gebhart 2005; Srivastava 2020; Kesarwani 2018; Ali & Ali 2013	Directly parameterised by measurable visibility V . Well-validated against fog chamber data.	Does not capture rapid temporal fluctuations. Kim model is empirical; accuracy degrades for $V < 50$ m.
h_{GG} (Gamma-Gamma turbulence)	Random multiplicative irradiance fading from atmospheric scintillation. Modelled as Gamma-Gamma distributed random variable with parameters α, β derived from Rytov variance.	Nistazakis 2009; Pernice 2016; Alshaer 2021; García-Zambrana 2012	Valid from weak to strong turbulence. Closed-form pdf enables analytical BER/outage derivations. Experimentally validated.	α, β estimation from Rytov variance is approximate. Does not capture temporal correlation directly (addressed by Markov chain extensions: Luo & Dang 2024).
h_{point} (Pointing error fading)	Random geometric loss from beam-detector misalignment due to platform vibration/sway. Modelled as a function of beam radius w_z , aperture radius a , and jitter σ_s .	García-Zambrana 2012; Bhatnagar 2015; Borah & Voelz 2009; Verma 2017	Physically motivated. Tractable closed-form expressions available. Captures building-sway effects critical for urban FSO.	Assumes zero-mean Gaussian jitter. Non-symmetric or drift-induced pointing errors need extensions.

5 Final Recommendations

In summary, the reviewed literature supports a composite atmospheric channel consisting of Gamma-Gamma turbulence, visibility-based attenuation, and pointing-error fading. Based on implementation complexity and literature support, the recommended starting point is the development of a Gamma-Gamma turbulence model and Kim attenuation module, followed by

Table 5: Four-phase implementation plan

Phase	Objective	Inputs	Outputs	Deliverables
Phase 1: Turbulence baseline	Implement Gamma-Gamma turbulence; compute Rytov variance for Delhi C_n^2 and target link distances	C_n^2 range (10^{-13} – 10^{-15} m $^{-2/3}$), link length L , wavelength $\lambda = 1550$ nm	Rytov variance σ_R^2 ; α, β parameters; GG random variate h_{GG} ; scintillation index SI	MATLAB function: <code>GG_turbulence(Cn2, L, lambda)</code> . Validated against Nistazakis 2009 analytical curves.
Phase 2: Atmospheric attenuation	Implement Kim model (primary) and Al-Naboulsi (secondary) for Delhi seasonal visibility ranges	Visibility V (seasonal distribution), wavelength λ , link distance L	Attenuation coefficient β_{att} (dB/km); deterministic path loss h_{atm} ; seasonal link-budget table	MATLAB function: <code>kim_attenuation(V, lambda, L)</code> . Validation against Kesarwani 2018 results for Delhi fog scenarios.
Phase 3: Pointing error fading	Add Gaussian jitter misalignment block; characterise sensitivity to w_z, a, σ_s	Beam radius w_z , aperture radius a , jitter std σ_s (estimated from building vibration spec.)	Pointing loss fading variate h_{point} ; composite irradiance pdf with turbulence	MATLAB function: <code>pointing_loss(wz, a, sigma_s)</code> . BER vs. jitter curves reproducing García-Zambrana 2012 Fig. 4–6.
Phase 4: SNS-TFQKD integration	Map composite channel transmittance $\eta = h_{total}$ to QKD performance metrics; implement prefixed-threshold selection (Yu 2022)	$h_{atm}, h_{GG}, h_{point}$ outputs from Phases 1–3; detector efficiency η_d , dark count rate d_c , excess noise ε	QBER, detection gain Q , raw key rate R_{raw} , secret key rate SKR vs. link distance and visibility	MATLAB SNS-TFQKD simulator with atmospheric front-end from Phases 1–3. Comparison against Yu 2022 Fig. 2–4 as validation reference.

pointing-error integration and SNS-TFQKD performance evaluation. This phased approach ensures that the most critical channel impairments are captured early, while more refined temporal and site-specific effects can be incorporated in later stages as validated experimental data become available.

Table 6: Summary of recommendations for implementation

Question	Recommendation
1. Which turbulence model to implement first?	Gamma-Gamma (GG) distribution parameterised by α, β from Rytov variance σ_R^2 . Valid across all turbulence regimes. Experimentally validated by Pernice 2016; used by Alshaer 2021 for QKD-FSO. Log-Normal is <i>not</i> the default; use only as a weak-turbulence check.
2. Which attenuation model to implement first?	Kim model. Simplest visibility-based formulation; appears directly in both Delhi FSO papers (Srivastava 2020, Kesarwani 2018); validated against fog chamber data (Gebhart 2005; Saeed 2023). Upgrade to Al-Naboulsi when fog type can be classified.
3. Which pointing error model to implement first?	Gaussian jitter misalignment model of García-Zambrana et al. (2012). Parameters: beam radius w_z , aperture radius a , jitter std σ_s . Closed-form pdf; directly usable in BER and outage derivations.
4. Most important papers for implementation?	García-Zambrana 2012 (pointing+GG BER); Nistazakis 2009 (GG FSO capacity); Alshaer 2021 (QKD-FSO with GG+pointing); Yu 2022 (free-space SNS-TFQKD with PDTTC); Kesarwani 2018 / Srivastava 2020 (Delhi calibration).
5. Best-supported channel architecture?	Composite multiplicative model: $h_{\text{total}} = h_{\text{atm}} \times h_{\text{GG}} \times h_{\text{point}}$. Supported by García-Zambrana 2012, Alshaer 2021, and the Elicit synthesis report. Modular structure allows piecewise validation.
6. Implement immediately (Phase 1–2)?	Gamma-Gamma turbulence generator (Phase 1) and Kim attenuation block (Phase 2). These two components provide the dominant SNR-degradation effects for Delhi links and enable credible QKD link-budget estimates from day one.
7. Postpone to future work?	Finite-state Markov channel modelling for temporal correlations (Luo & Dang 2024); non-Gaussian pointing drift models; background solar noise as a standalone atmospheric model; experimental C_n^2 measurements at Delhi sites; and integration with OptiSystem for photonic simulation.